Optimization Problem: Storage Arbitrage under Net Metering

1 Energy Arbitrage Lab: Objective, Constraints, and Outcomes

The objective of the energy arbitrage lab is to determine the optimal charging and discharging schedule of a behind-the-meter battery in order to minimize the household electricity cost over a finite time horizon. The model considers time-varying electricity prices, household net load, battery efficiency losses, and net-metering compensation. The battery is operated behind the meter, meaning that it modifies the household net exchange with the grid.

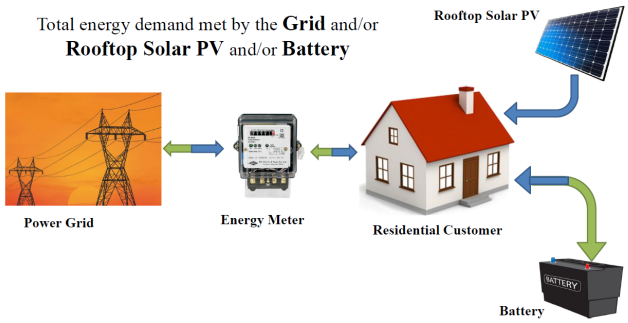

Figure 1: Behind-the-meter PV and battery system.

Two formulations are considered: a simplified *storage-only* arbitrage model and *storage-with-load* model under net metering.

1. Storage-only arbitrage model

This formulation optimizes the battery schedule based only on electricity prices, without explicitly considering household demand or renewable generation. The battery charges when prices are low and discharges when prices are high.

The optimization problem is [1]

$$\min_{x,z} \sum_{k=1}^{N} z_k \tag{1}$$

subject to

$$z_k \geq \frac{\text{real}_k}{\eta_{ch}} x_k, \qquad k = 1, \ldots, N, \tag{2}$$

$$z_k \geq \kappa * \text{real}_k * \eta_{dis}\, x_k, \qquad k = 1, \ldots, N, \tag{3}$$

$$\sum_{i=1}^{k} x_i \leq b_{\max} - b_0, \qquad k = 1, \ldots, N, \tag{4}$$

$$-\sum_{i=1}^{k} x_i \leq b_0 - b_{\min}, \qquad k = 1, \ldots, N, \tag{5}$$

$$\Delta_{\min} * h \leq x_k \leq \Delta_{\max} * h, \qquad k = 1, \ldots, N. \tag{6}$$

Here, x_k is the battery charging/discharging decision and z_k is an auxiliary variable representing the transaction cost. The constraints enforce piecewise linear cost, battery state-of-charge limits, and charge/discharge rate bounds.

2. Storage with load under net metering

A more complete formulation includes the household net load [1]

$$z_t = d_t - r_t, \tag{7}$$

where d_t is inelastic demand and r_t is renewable generation. The selling price is modeled as a fraction of the buying price:

$$p_s(t) = \kappa * p_b(t), \qquad 0 \leq \kappa \leq 1. \tag{8}$$

The battery contribution at the meter is

$$s_t = \frac{1}{\eta_{ch}}[x_t]^+ - \eta_{dis}[x_t]^-, \tag{9}$$

so that the net grid exchange becomes

$$\ell_t = z_t + s_t. \tag{10}$$

The electricity cost under net metering is

$$C_t(\ell_t) = p_b(t)[\ell_t]^+ - p_s(t)[-\ell_t]^+. \tag{11}$$

To preserve linearity, an epigraph variable u_t is introduced and the problem is written as [1]

$$\min_{x,u} \sum_{t=1}^{T} u_t \tag{12}$$

subject to

$$\frac{p_b(t)}{\eta_{ch}} x_t - u_t \leq -z_t p_b(t), \tag{13}$$

$$p_b(t)\eta_{dis} x_t - u_t \leq -z_t p_b(t), \tag{14}$$

$$\frac{p_s(t)}{\eta_{ch}} x_t - u_t \leq -z_t p_s(t), \tag{15}$$

$$p_s(t)\eta_{dis} x_t - u_t \leq -z_t p_s(t), \tag{16}$$

for all $t = 1, \ldots, T$, together with

$$\sum_{\tau=1}^{t} x_\tau \leq b_{\max} - b_0, \tag{17}$$

$$-\sum_{\tau=1}^{t} x_\tau \leq b_0 - b_{\min}, \tag{18}$$

$$\Delta_{\min} * h \leq x_t \leq \Delta_{\max} * h. \tag{19}$$

This formulation accounts for household demand, renewable generation, import/export pricing, and battery efficiency losses. It therefore represents the full behind-the-meter arbitrage problem under net metering.

Performance metrics and outcomes

The main performance metrics are the baseline electricity bill without storage,

$$C_{\text{nom}} = \sum_{t=1}^{T} \left(p_b(t)[z_t]^+ - p_s(t)[-z_t]^+ \right), \tag{20}$$

the optimized bill with storage,

$$C_{\text{bat}} = \sum_{t=1}^{T} \left(p_b(t)[\ell_t]^+ - p_s(t)[-\ell_t]^+ \right), \tag{21}$$

and the arbitrage gain,

$$\Pi = C_{\text{nom}} - C_{\text{bat}}. \tag{22}$$

The main outcomes of the lab are the optimal battery schedule, the state-of-charge profile, and the total cost reduction. In general, the battery charges during low-price periods and discharges during high-price periods, thereby lowering the electricity bill. The state-of-charge trajectory confirms that battery operation respects the energy limits. A sensitivity analysis with respect to κ further shows that the economic value of storage strongly depends on export compensation: larger κ values make discharging more profitable and increase the achievable savings.

2 Conceptual Understanding Questions

1. **Behind-the-meter meaning**
   Explain what *behind-the-meter assets* mean. What signals does the grid see before and after adding the battery?

2. **Net load definition**
   If
   $$z_t = d_t - r_t,$$
   what do
   $$z_t > 0 \quad \text{and} \quad z_t < 0$$
   mean physically?

3. **Why efficiencies matter**
   Why do we write the battery contribution as
   $$s_t = \frac{1}{\eta_{ch}}[x_t]^+ - \eta_{dis}[x_t]^-$$
   instead of
   $$s_t = x_t\,?$$

4. **Buying vs selling price**
   What does
   $$\kappa$$
   represent? Why must
   $$\kappa \leq 1$$
   under typical net-metering assumptions?

References

[1] M. U. Hashmi, A. Mukhopadhyay, A. Bušić, J. Elias, and D. Kiedanski, "Optimal storage arbitrage under net metering using linear programming," in *2019 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, 2019, pp. 1–7.